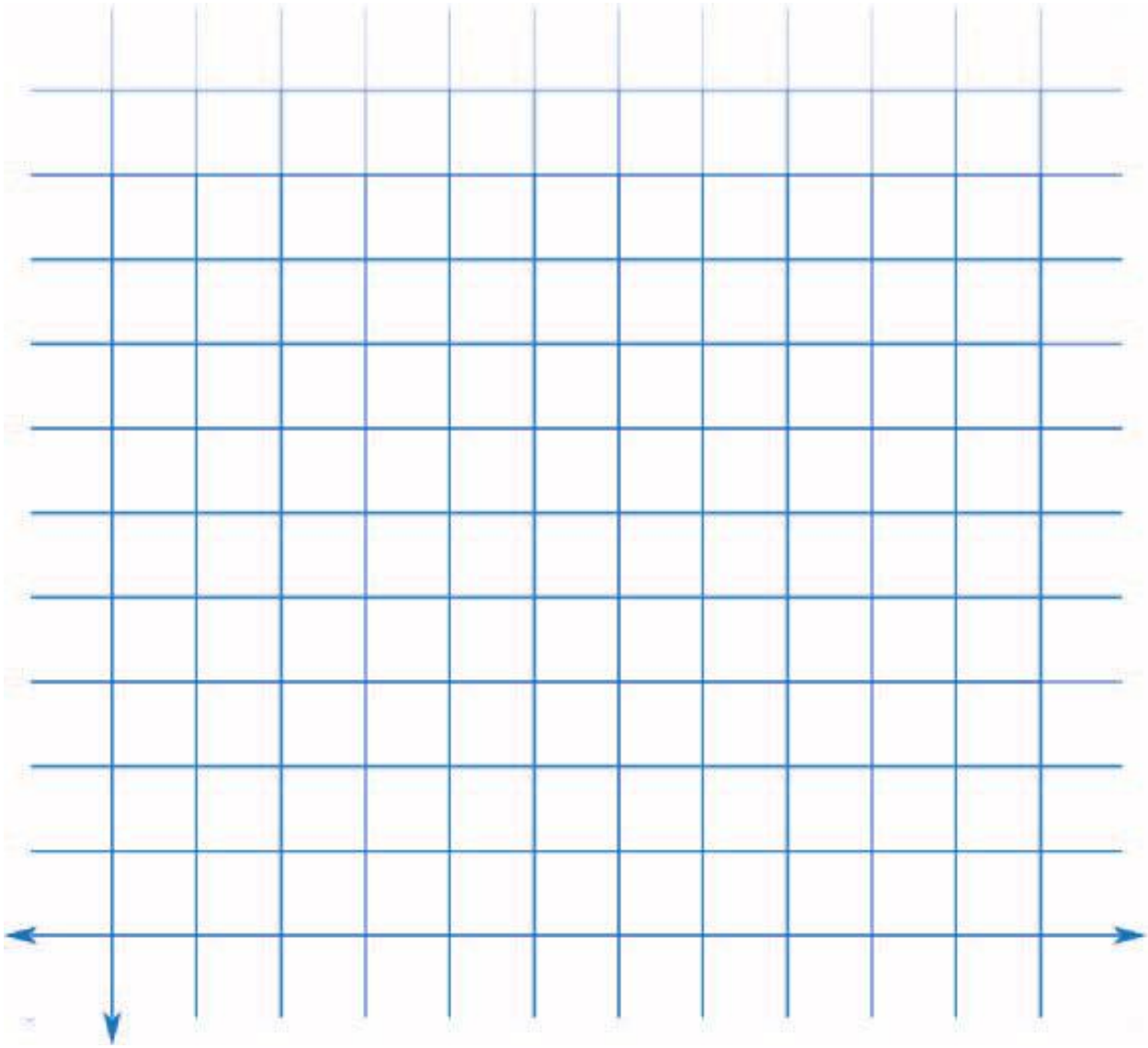
Momentum

Carolina Distance Learning
Investigation Manual



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Overview

In the following activities, students will investigate how the law of conservation of momentum applies to collisions. Students will perform a series of experiments and record data on mass, distance, and time, which are used to calculate the velocity and momentum of moving components of a system before and after a collision.

Objectives

- Observe collisions between two dynamic carts
- Determine how differences in cart mass affect the collisions
- Apply the law of conservation of momentum to collisions

Time Requirements

Preparation.....15 minutes
Activity 1 .....25 minutes
Activity 2 .....25 minutes
Activity 3 .....25 minutes

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Background

Momentum is a vector. The momentum **p** of an object is the product of its mass **m** (in kg) times its velocity **v** (in m/s), as in the following equation:

$$\boldsymbol{p} = m\boldsymbol{v}$$

The law of conservation of momentum holds that the amount of momentum in a system must remain constant if that system is not affected by external forces.

The law of conservation of momentum is often applied to analyzing collisions, because if there are no external forces on the system, the total momentum of all the components of the system before the collision must equal the total momentum of those components after the collision.

The game of pool offers a good example of the law of conservation of momentum. Consider a cue ball that strikes a stationary pool ball (**Figure 1**). The cue ball starts with a mass $\mathbf{m_1}$ and a velocity $\mathbf{v_1}$. The stationary ball starts with a mass $\mathbf{m_2}$ and a velocity $\mathbf{v_2}$, After the collision, the two balls move at different velocities than those before the collision, but the total momentum of the system (i.e., the sum of the momentums of each ball) will be the same before and after the collision.

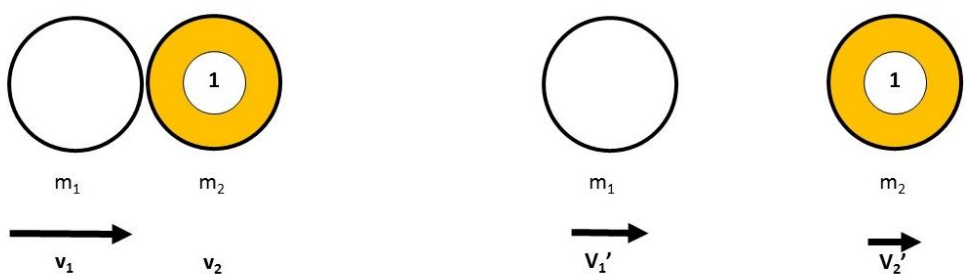

Figure 1

The application of the law of conservation to this example is demonstrated by the following equation:

$$\boldsymbol{p}_1 + \boldsymbol{p}_2 = \boldsymbol{p}_1{}' + \boldsymbol{p}_2{}'$$
$$m_1\boldsymbol{v}_1 + m_2\boldsymbol{v}_2 = m_1\boldsymbol{v}_1{}' + m_2\boldsymbol{v}_2{}'$$

Note: By convention, momentum and velocity after the collision are indicated with a "prime" symbol ('). **v'** is read "v prime".

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Consider the collision of two billiard balls. The cue ball has a mass $\mathbf{m_1}$ of 0.17 kg and an initial velocity $\mathbf{v_1}$ of 3 m/s. The cue ball strikes a stationary pool ball that has a mass $\mathbf{m_2}$ of 0.17 kg and a $\mathbf{v_2}$ of 0 m/s. After the collision, the cue ball has a velocity $\mathbf{v_1'}$ of 1.2 m/s. What is the final velocity of the other ball?

The law of conservation of momentum can be applied using the following equations:

$$m_1\boldsymbol{v}_1 + m_2\boldsymbol{v}_2 = m_1\boldsymbol{v}_1{}' + m_2\boldsymbol{v}_2{}'$$
$$(0.17\ kg)\left(3\frac{m}{s}\right) + (0.16\ kg)\left(0\frac{m}{s}\right) = (0.17\ kg)\left(1.2\frac{m}{s}\right) + (0.16\ kg)(v_2')$$
$$0.51\ kg \cdot \frac{m}{s} = 0.204\ kg\ \cdot \frac{m}{s} + (0.16\ kg)(v_2')$$
$$0.306\ kg\ \cdot \frac{m}{s} = (0.16\ kg)(v_2{}')$$
$$v_2' = 1.9\ m/s$$

The momentum before the collision, represented by the left side of the equation, should equal the momentum after the collision, represented by the right side of the equation.

$$(0.17\ kg)\left(3\frac{m}{s}\right) + (0.16\ kg)\left(0\frac{m}{s}\right) = (0.17\ kg)\left(1.2\frac{m}{s}\right) + (0.16\ kg)(1.9\frac{m}{s})$$
$$0.51\ kg\ \cdot \frac{m}{s} = 0.51\ kg\ \cdot \frac{m}{s}$$

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Materials

Included in the central materials kit:

Tape measure	1
String	1
Pocket scale	1
Washers/weights	30
Protractor	1

Included in the Mechanics Module materials kit:

Foam board	1
Dynamic carts	2

Needed but not supplied:

Book	1
Calculator or computer	1
Masking tape	1
Permanent marker	1
Optional: Smartphone/tablet or other digital recording device	

Reorder information: Replacement supplies for the DL Physics Momentum investigation (kit 580404) can be ordered from Carolina Biological Supply Company. Call 1-800-334-5551 to order.

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Safety

Safety goggles should be worn at all times during these activities.

Read all the instructions for this laboratory activity before beginning. Follow the instructions closely and observe established laboratory safety practices, including the use of appropriate personal protective equipment (PPE) described in the Safety and Procedure section.

Safety goggles should be worn during these activities, which involve the movement and acceleration of objects. Take care during the execution of these activities to avoid injuring hands and fingers or feet and toes through contact with moving or falling objects. Make sure the area is clear of pets, children, and breakable objects.

Do not eat, drink, or chew gum while performing this activity. Wash your hands with soap and water before and after performing the activity. Clean up the work area with soap and water after completing the investigation. Keep pets and children away from lab materials and equipment.

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Alternate Methods for Collecting Data using Digital Devices.

Much of the uncertainty in these experiments arises from human error in measuring the times of events. Some of the time intervals are very short, which increases the effect of human error due to reaction time.

Observing the experiment from a good vantage point that removes parallax errors and recording measurements for multiple trials helps to minimize error, but using a digital device as an alternate method of data collection may further minimize error.

Many digital devices, smart phones, tablets, etc. have cameras and software that allow the user to pause or slow down the video.

If you film the experiment against a scale, such as a tape measure, you can use your video playback program to record position and time data for the carts. This can provide more accurate data and may eliminate the need for multiple trials.

If the time on your device's playback program is not sufficiently accurate, some additional apps may be available for download.

Another option is to upload the video to your computer. Different video playback programs may come with your operating system or software suite or may be available for download.

Some apps for mobile devices and computer programs available for download are listed below, with notes about their features.

Hudl Technique: http://get.hudl.com/products/technique/

- *iPhone/iPad and Android*
- *FREE*
- *Measures times to the hundredth-second with slow motion features*

QuickTime http://www.apple.com/quicktime/download/

- Free
- Install on computer
- 30 frames per second
- Has auto scrubbing capability

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Preparation

Watch the following video for an overview of how to assemble your momentum carts.
http://bcove.me/qcsf1m6i

1. Attach a rubber stopper to the front of one cart with the steel nut provided. This is Cart A.
2. Secure a steel spring with a rubber stopper and attach it to the front bumper of the other cart. This is Cart B.
3. On a level surface such as a table, set up a ramp at an angle of 10-15° using the foam board and one or more books (**Figure 2**). Use a protractor to measure the angle.

> **Note:** The foam board may bend in the middle as the cart rolls down the ramp, especially in later experiments when mass is added. It may be necessary to add one or more books beneath the foam board for support. The angle of the board should allow the cart to transition from the board to the table without its front end or rubber stopper contacting the table.

4. Secure the tape measure along the level surface where the carts will move. The starting point of the tape measure should begin at the lower end of the foam board.
5. Using pieces of masking tape, mark 10-cm increments along length of the table.
6. Mark the piece of tape 20 cm from the foam board as the **start point**.
7. Mark the piece of tape 70 cm from the foam board as the **end point**.
8. Place Cart B behind the end point so that the steel spring is behind the piece of tape.
9. Place a book at the other end of Cart B so that the steel spring is behind the end point when Cart B makes contact with the book. (See **Figure 2**)
10. Place Cart A at the top of the foam board so that the back end of the cart is even with the end of the board. Use a permanent marker or piece of tape to make a line on the board indicating the **release point**. The front of the cart should be behind this line at the start of every trial.

> **Note:** See Figure 2 for a visual depiction of the experimental set-up.

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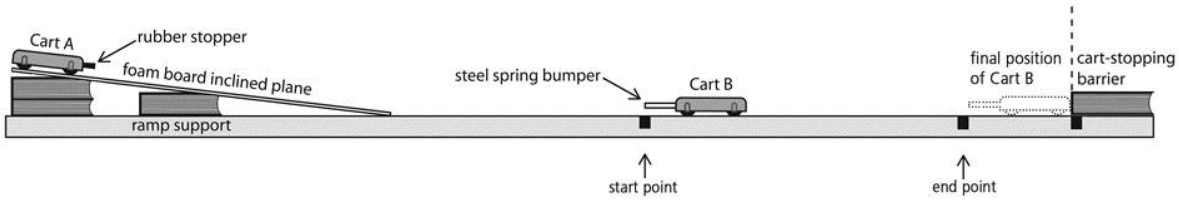

Figure 2

Activity 1. Elastic Collision with Equal Masses

1. Measure the mass of Cart A and Cart B and record the data in the appropriate section of Tables 1A, 1B, and 1C.

2. You will now measure the initial velocity $\mathbf{v_A}$ of Cart A. Remove Cart B, and place Cart A behind the release point at the raised end of the foam board.
3. Release Cart A, and allow Cart A to travel down the ramp onto the level surface.
4. Measure the time it takes for Cart A to travel from the start point to the end point. Start the timer when the back end of Cart A crosses the start point, and stop the timer when the front end of Cart A strikes the book. The distance for this trial will be 50 cm.
5. Repeat this process for three trials, and record the data in **Data Table 1A.**
6. You will now determine the final velocity $\mathbf{v_A'}$ of Cart A. Place Cart B on the level surface. The spring should be at the start point.
7. Place Cart A at the raised end of the foam board behind the release point.
8. Release Cart A, and allow Cart A to impact Cart B. It is important that the impact is straight and that both carts travel in a straight line along Cart A's original direction after the collision.
9. Start timing when Cart A collides with Cart B and until Cart A stops moving.
10. Observe the distance Cart A travels after the collision. If the cart does not move the distance is zero, and if the cart moves backward the distance is negative.
11. Record the distance Cart A moves after the collision and record this distance in **Data Table 1B**.
12. Repeat this process for three trials. Record the time for each trial in **Data Table 1B**.
13. You will now measure the final velocity $\mathbf{v_B'}$ of Cart B. Place Cart A at the raised end of the foam board behind the release point.
14. Release Cart A, and allow Cart A to impact Cart B. It is important that the impact is straight and that both carts travel in a straight line along Cart A's original direction after the collision.
15. Record the time for Cart B to travel from the start point to the end point. This distance is 50 cm. Record this value in **Data Table 1C**.
16. Repeat this process for three trials, and record the data in Table 1C.
17. Complete **Data Tables 1A, 1B, and 1C** by calculating the average velocity of the carts before and after the collision.

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Data Table 1A. Cart A before collision.

Cart A mass, m_1 (kg)	Distance, d (m)	Time, t (s)	Average time, t (s)	Velocity = d/t (m/s) $\mathbf{v_A}$

Data Table 1B. Cart A after collision.

Cart A mass, m_1 (kg)	Distance, d (m)	Time, t (s)	Average time, t (s)	Velocity = d/t (m/s) $\mathbf{v_A'}$

Data Table 1C. Cart B after collision.

Cart B mass, m_2 (kg)	Distance, d (m)	Time, t (s)	Average time, t (s)	Velocity = d/t (m/s) $\mathbf{v_B'}$

Calculations for Activity 1. Elastic Collision with Equal Masses.

Apply the law of conservation of momentum to the two-cart system by calculating the momentum before and after the collision.

1. Calculate the momentum of the system before the collision (the left side of the equation) and after the collision (the right side of the equation).

2. Calculate the percent difference between the two values.

3. Explain any difference in the values before and after the collision.

$$m_A \boldsymbol{v}_A + m_B \boldsymbol{v}_B = m_A \boldsymbol{v}_A' + m_B \boldsymbol{v}_B'$$

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Momentum before the collision = $m_A \boldsymbol{v}_A + m_B \boldsymbol{v}_B$

Momentum after the collision = $m_A \boldsymbol{v}_A{}' + m_B \boldsymbol{v}_B{}'$

$$percent\ difference = \left| \frac{first\ value - second\ value}{\left(\frac{first\ value + second\ value}{2}\right)} \right| \ x\ 100\%$$

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Activity 2. Elastic Collision: Mass Added to Cart A

Repeat the procedure with mass added to Cart A.

1. Add five large washers to Cart A.
2. Using the electronic pocket balance, measure the mass of Cart A with the additional washers. Record this value in **Data Table 2A and 2B**.
3. Using the electronic pocket balance, measure the mass of empty Cart B. Record this value in **Data Table 2C**.
4. You will now determine the initial velocity $\mathbf{v_A}$ of Cart A. Remove Cart B from the surface, and place Cart A behind the release point at the raised end of the foam board.
5. Release Cart A, and allow Cart A to travel down the ramp onto the level surface.
6. Measure the time it takes for Cart A to travel from the start point to the end point. Start the timer when the back end of Cart A crosses the start point, and stop the timer when the front end of Cart A strikes the book. (The distance for this trial will be 50 cm.)
7. Repeat this process for three trials, and record the data in **Data Table 2A**.
8. You will now determine the final velocity vA' of Cart A. Place Cart B on the level surface. The spring should be at the start point.
9. Place Cart A at the raised end of the foam board behind the release point.
10. Release Cart A, and allow Cart A to impact Cart B. It is important that the impact is straight and that both carts travel in a straight line along Cart A's original direction after the collision.
11. Start timing when Cart A collides with Cart B and until Cart A stops moving.
12. Observe the distance Cart A travels after the collision. If the cart does not move the distance is zero, and if the cart moves backward the distance is negative.
13. Use a piece of tape to mark a position on the table 40 cm from the end of the foam board. Record a distance of 20 cm (40 cm-20 cm) in **Data Table 2B** as the distance Cart A travels after the collision.

> **Note:** 40 cm is a suggested end point. Variables such as the angle of the ramp and the type of surface you use may require you to select a different distance in order to calculate the final velocity, $\mathbf{v_A'}$, for Cart A. The distance should be about ½ to 2/3 the total distance Cart A travels after the collision so that the velocity of Cart A over this distance can be considered constant. Measure the distance to the end point you select from the start point at 20 cm. Record this distance in Table 2B. Remember if Cart A reverses direction, $\mathbf{v_A'}$ will be negative.

14. Measure the time for Cart A's movement after the collision for three trials. Record the time for each trial, and record the data in **Data Table 2B**.
15. You will now find the final velocity $\mathbf{v_B'}$ of Cart B. Place Cart A at the raised end of the foam board behind the release point.

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16. Release Cart A, and allow Cart A to impact Cart B. It is important that the impact is straight and that both carts travel in a straight line along Cart A's original direction.
17. Record the time it takes for Cart B to travel from the start point to the end point.
18. Repeat this process for three trials, and record the data in **Data Table 2C**.
19. Measure the distance from the start point to the end point. (This distance should be 50 cm) Record this value in **Data Table 2C**.
20. Complete **Data Tables 2A, 2B, and 2C** by calculating the average velocity of the carts before and after the collision.

Data Table 2A. Cart A before collision.

Cart A mass, m_1 (kg)	Distance, d (m)	Time, t (s)	Average time, t (s)	Velocity = d/t (m/s) $\mathbf{v_A}$

Data Table 2B. Cart A after collision.

Cart A mass, m_1 (kg)	Distance, d (m)	Time, t (s)	Average time, t (s)	Velocity = d/t (m/s) $\mathbf{v_A'}$

Data Table 2C. Cart B after collision.

Cart B mass, m_2 (kg)	Distance, d (m)	Time, t (s)	Average time, t (s)	Velocity = d/t (m/s) $\mathbf{v_B'}$

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Calculations for Activity 2. Elastic Collision: Mass Added to Cart A.

Apply the law of conservation of momentum to the two-cart system by calculating the momentum before and after the collision.

1. Calculate the momentum of the system before the collision (the left side of the equation) and after the collision (the right side of the equation).

2. Calculate the percent difference between the two values.

3. Explain any difference in the values before and after the collision.

$$m_A\boldsymbol{v}_A + m_B\boldsymbol{v}_B = m_A\boldsymbol{v}_A' + m_B\boldsymbol{v}_B'$$

Momentum before the collision = $m_A\boldsymbol{v}_A + m_B\boldsymbol{v}_B$

Momentum after the collision = $m_A\boldsymbol{v}_A' + m_B\boldsymbol{v}_B'$

$$percent\ difference = \left| \frac{first\ value - second\ value}{\left(\frac{first\ value + second\ value}{2}\right)} \right| x\ 100\%$$

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Activity 3. Elastic Collision: Mass Added to Cart B

Repeat the procedure with mass added to Cart B.

Note: In this activity, Cart A may have a negative velocity after the collision. To allow enough space to measure the velocity of Cart A, change the **start point** to 35 cm from the end of the foam board.

1. Place 11 large washers in Cart B and five small washers in Cart A.
2. Using the electronic pocket balance, measure the mass of Cart B with the 11 washers. Record this value in **Data Table 3C**.

> **Note:** For this activity, the mass of the target cart, Cart B, should have more Mass than Cart A. In this scenario, Cart A will likely reverse direction after the collision. If the mass of Cart A is too small, it will move to fast to record an accurate time. The numbers of washers in step 1 are suggestions. You may need to adjust the number of washers in each Cart to get an accurate measurement for time for each cart.

3. Using the electronic balance, measure the mass of Cart A with the five small washers. Record this value in **Data Table 3A and 3B**.
4. Using a piece of tape, mark a point on the level surface 35 cm from the lower end of the foam board. This will be the new **start point**.

> **Note:** In this scenario, Cart A will likely reverse direction after the collision. Moving the start point, where the collision will occur, farther from the foam board should provide adequate space for Cart A to move after the collision so that a time can be recorded and a velocity can be calculated. Remember if Cart A reverses direction after the collision then $\mathbf{v_A'}$ will be negative.

5. You will now determine the initial velocity $\mathbf{v_A}$ of Cart A. Remove Cart B, and place Cart A behind the release point at the raised end of the foam board.
6. Release Cart A, and allow Cart A to travel down the ramp onto the level surface.
7. Measure the time it takes for Cart A to travel from the start point to the end point. Start the timer when the back end of cart a crosses the start point, and stop the timer when the front end of Cart A strikes the book. The distance for this trial will be 35 cm (70 cm – 35 cm).
8. Repeat this process for three trials, and record the data in **Data Table 3A**.
9. You will now find the final velocity $\mathbf{v_A'}$ of Cart A. Place Cart B on the level surface at the start point.
10. Place Cart A at the raised end of the foam board behind the release point.
11. Release Cart A, and allow Cart A to impact Cart B. It is important that the impact is straight and that both carts travel in a straight line after the collision.
12. Observe the distance and direction Cart A travels after the collision. If Cart A does not move after the collision, remove a washer from Cart A and repeat the previous step.

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13. Once Cart A and B move consistently after the collision, measure a point 10 cm from the start point. This will be the end position for Cart A. Record this distance in **Data Table 3B**.

> **Note:** 10 cm is a suggested end point. Variables such as the angle of the ramp and the type of surface you use may require you to select a different distance in order to calculate the final velocity, $\mathbf{v_A'}$, for Cart A. The distance should be about ½ to 2/3 the total distance Cart A travels after the collision so that the velocity of Cart A over this distance can be considered constant. Measure the distance to the end point you select from the start point at 35 cm. Record this distance in Table 3B. Remember if Cart A reverses direction, $\mathbf{v_A'}$ will be negative.

14. Record the time for Cart A to travel from the start point (where the collision occurs) to the end point for Cart A, 10 cm from the start point.
15. Repeat this process for three trials, and record the data in **Data Table 3B**.
16. You will now find the final velocity $\mathbf{v_B'}$ of Cart B. Measure the distance from the start point to the end point. This distance should be 35 cm (70 cm -35 cm). Record this distance in Table 3C.
17. Place Cart A at the raised end of the foam board behind the release point.
18. Release Cart A, and allow Cart A to impact Cart B. It is important that the impact is straight and that both carts travel in a straight line along Cart A's original direction after the collision.
19. Record the time it takes for Cart B to travel from the start point to the end point.
20. Repeat this process for three trials, and record the data in **Data Table 3C**.
21. Complete **Data Tables 3A, 3B, and 3C** by calculating the average velocity of the carts before and after the collision.

Data Table 3A. Cart A before collision.

Cart A mass, m_1 (kg)	Distance, d (m)	Time, t (s)	Average time, t (s)	Velocity = d/t (m/s) $\mathbf{v_A}$

Data Table 3B. Cart A after collision.

Cart A mass, m_1 (kg)	Distance, d (m)	Time, t (s)	Average time, t (s)	Velocity = d/t (m/s) $\mathbf{v_A'}$

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Data Table 3C. Cart B after collision.

Cart B mass, m_2 (kg)	Distance, d (m)	Time, t (s)	Average time, t (s)	Velocity = d/t (m/s) $\mathbf{v_B'}$

Calculations for Activity 3. Elastic Collision: Mass Added to Cart B.

Apply the law of conservation of momentum to the two-cart system by calculating the momentum before and after the collision.

1. Calculate the momentum of the system before the collision (the left side of the equation) and after the collision (the right side of the equation.)

2. Calculate the percent difference between the two values.

3. Explain any difference in the values before and after the collision.

$$m_A\boldsymbol{v}_A + m_B\boldsymbol{v}_B = m_A\boldsymbol{v}_A' + m_B\boldsymbol{v}_B'$$

Momentum before the collision = $m_A\boldsymbol{v}_A + m_B\boldsymbol{v}_B$

Momentum after the collision = $m_A\boldsymbol{v}_A' + m_B\boldsymbol{v}_B'$

$$percent\ difference = \left|\frac{first\ value - second\ value}{\left(\frac{first\ value + second\ value}{2}\right)}\right| x\ 100\%$$

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